

Assignment - 3

CSL 2050 - Pattern Recognition & Machine Learning

Max. Marks: 50

Submission Instructions:

- Students must submit one Jupyter notebook (.ipynb) file containing:
 - Executable code
 - Visible outputs (tables, figures, metrics)
 - Brief comments or markdown explanations where required
- File naming format : **RollNo.-Name.ipynb** example, 21CS045-AmitSharma.ipynb
- Do Not submit PDFs, word files, or screenshots.
- Late submissions will be penalized as per course policy.
- Date of last submission: **1st April 2026, 11:59PM.**

Part A: Multi-Layer Perceptron (from scratch)

Dataset: XOR dataset

https://www.kaggle.com/datasets/bipinmaharjan/xor-dataset?select=Xor_Data_set.csv

Tasks:

1. Implement a single neuron. [5]
 - a. Write a function for perceptron with inputs as weights, bias and input vector and output as predicted value.
 - b. Implement the following: (i) weighted sum computation (ii) Activation function
2. Implement a loss function such as mean squared error (MSE) [5]

3. Using the perceptron function as building block [5]
 - a. Create a network with: (i) input layer (2 neurons) (ii) hidden layer (2-3 neurons) (iii) output layer (1 neuron)
 - b. Use activation function
 - c. Implement forward propagation
4. Backpropagation [5]
 - a. Compute gradients manually
 - b. Update weights using gradient descent
 - c. Train the MLP on XOR dataset
5. Evaluation [5]
 - a. Show prediction before and after training
 - b. Show loss trend over epochs and plot loss vs epochs

Part B: Convolutional Neural Networks (use libraries such as PyTorch)

Dataset: Fashion MNIST (call directly using PyTorch)

Tasks:

1. Load the dataset using PyTorch. [5]
 - a. Create dataloader for training and testing
 - b. Visualize few samples
2. Build simple CNN architecture with convolution layers to be equal to 2 along with structures such as ReLU activation function, max-pooling, flatten layer, etc. Use different learning rate techniques such as constant and annealing. Vary the following parameters
 - a. Filter size: 3x3, 5x5
 - b. Zero and Same Padding
 - c. Stride: 1, 2 [10]
3. Train the CNN architecture with loss for classification. [5]

4. Evaluate model on test data for different architectures and compare the performances of each model with respect to different learning rate techniques. [5]